FIT PM12 twelve-channel test campaign

FIT hardware acceptance bench

November 13, 2026

1 Scope and recorded coverage

Board PM12-K7-3A17A1B402212223. Campaign 5ef2e2b4-c04a-4fad-b86f-a23be621d7a7.

Recorded status: INCOMPLETE. Missing tests are NOT MEASURED. Complete selected-protocol coverage is not a calibrated whole-board release.

Channel	cfd-monitor	cfd-dacs	gate-phase	tdc	adc-dc	adc-lines	integrator
1	NM	NM	NM	NM	NM	NM	NM
2	NM	NM	NM	NM	NM	NM	NM
3	NM	NM	NM	NM	NM	NM	NM
4	NM	NM	NM	NM	NM	NM	NM
5	NM	NM	NM	NM	NM	NM	NM
6	NM	NM	NM	NM	NM	NM	NM
7	NM	NM	NM	NM	NM	NM	NM
8	NM	NM	NM	NM	NM	NM	NM
9	NM	NM	NM	NM	NM	NM	NM
10	NM	NM	NM	NM	NM	NM	NM
11	NM	NM	NM	NM	NM	NM	NM
12	NM	NM	NM	NM	PASS	PASS	PASS

NM = not measured; INC = incomplete; ERR = acquisition error. No missing series is invented.

1.1 Board-level tests

Test	Status
Boundary scan and CPLD–FPGA connections	NOT_MEASURED
MCU, CPU–FPGA communication and peripheral IDs	NOT_MEASURED
I2C sensors, currents and voltages	NOT_MEASURED
HDMI digital link	NOT_MEASURED
Ethernet/bridge diagnostic	NOT_MEASURED

2 Fitted batches and fixture provenance

Batch attempt	PM channel	CFD label	Declared connection	Fitting
ch09-12-attempt001	12	CFD01	slot 7, RJ45 0	Identity/fitting confirmed
ch09-12-attempt001	11	CFD02	slot 7, RJ45 1	Identity/fitting confirmed
ch09-12-attempt001	10	CFD03	slot 7, RJ45 2	Identity/fitting confirmed
ch09-12-attempt001	9	CFD04	slot 7, RJ45 3	Identity/fitting confirmed

Fixture labels CFD01–CFD04 are provisional bench labels. Factory identities and photos remain unknown until supplied; a label is not proof of module identity. Declared cable routes must be checked physically.

Raw inputs and selected result IDs are retained in inputs/. Repeated measurements are never silently substituted.

3 Retained attempts and explicit selection

Result ID prefix	Test	Channel	Fitting	Status	Selected
result-b9126235	adc-dc	12	ch09-12-attempt001	PASS	yes
result-ee0613ea	adc-lines	12	ch09-12-attempt001	PASS	yes
result-217f5cb4	integrator	12	ch09-12-attempt001	PASS	yes

All attempts, including failures and rejected repeats, remain in the archived inputs. The coverage matrix uses only the explicitly selected attempt.

4 Selected measurement conditions and limits

Test	Physical channel	Confirmed stimulus connection
adc-dc	12	{”dac_channels”: [0, 1], ”stimulus_channel”: 12, ”usb_input”: ”J8”}
adc-lines	12	{”dac_channels”: [0, 1], ”stimulus_channel”: 12, ”usb_input”: ”J8”}
integrator	12	{”dac_channels”: [2, 3], ”stimulus_channel”: 12, ”usb_input”: ”J6”}

4.1 Common hardware configuration

Setting	Recorded value
clock_profile.adc_rate	2x reference
clock_profile.kasli_rate	3x reference
clock_profile.reference_mhz	40.0787
cpld_gate_mode	original INTA/INTB enabled, nomina l20MHz
kasli_build.ABI	CFD4
kasli_build.expected_bit_sha256	a71a08a258620b0e5a52c5d03aefd8c69040 c9d15e8de6391c484ac182b7173b
pm_build	c7c42d28

4.2 adc-dc

Setting	Recorded value
method	”direct-ADC-DC-v1”
stimulus_points_per_bank	257
commanded_range_V	[-1.2, 1.2]
common_mode_V	1.0
dac_channels	[0, 1]
fit_interval_V	[-0.8, 0.8]
fft_sample_rate_assumed_Hz	80157400
integrator_input_parked_V	[2.0480001, 1.02400005]

4.3 adc-lines

Setting	Recorded value
method	”adjacent-pair-states-v1”
stimulus_points_per_bank	257
commanded_range_V	[-1.2, 1.2]
common_mode_V	1.0
dac_channels	[0, 1]
fit_interval_V	[-0.8, 0.8]
fft_sample_rate_assumed_Hz	80157400
integrator_input_parked_V	[2.0480001, 1.02400005]

4.4 integrator

Setting	Recorded value
method	”signed-integration-pair-v1”
dac_channels	[2, 3]
base_V	[1.024, 2.048]
coarse_offsets_V	[-0.09999999999999987, -0.080000000000 00007, -0.05999999999999983, -0.040000 000000000036, -0.019999999999999796, -0.010000000000000009, 0.0, 0.009999999999 999787, 0.02000000000000024, 0.040000 000000000036, 0.060000000000000275, 0.08000000000000007, 0.10000000000000009]

Setting	Recorded value
fine_method	”26 points,0.4mV step around coarse estima te”
return_method	”two candidate revisits and +/-5mV offset s;near-zero <=2codes; signed response >4codes”

Commanded DAC voltages, whole connected chain; not calibrated ADC-only INL/noise or electrical continuity proof.

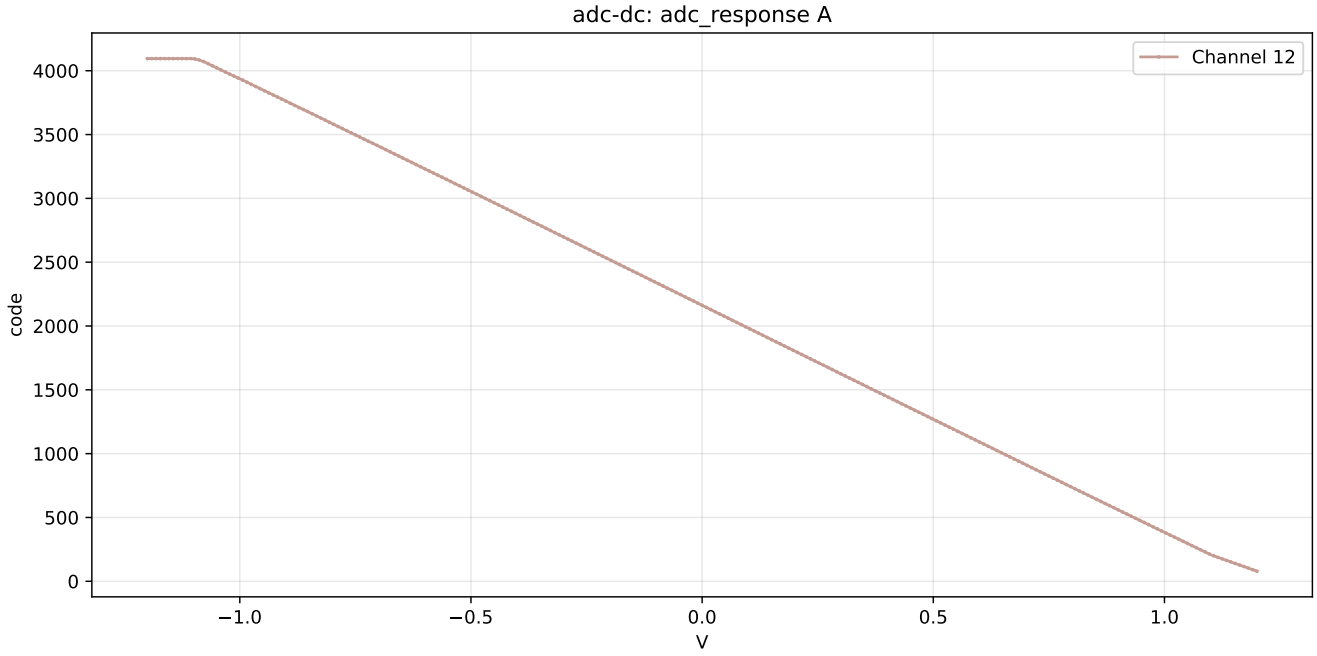
Effective differential whole-chain balance; waveform-based integration pair, not an absolute gate timestamp or calibrated opamp bias.

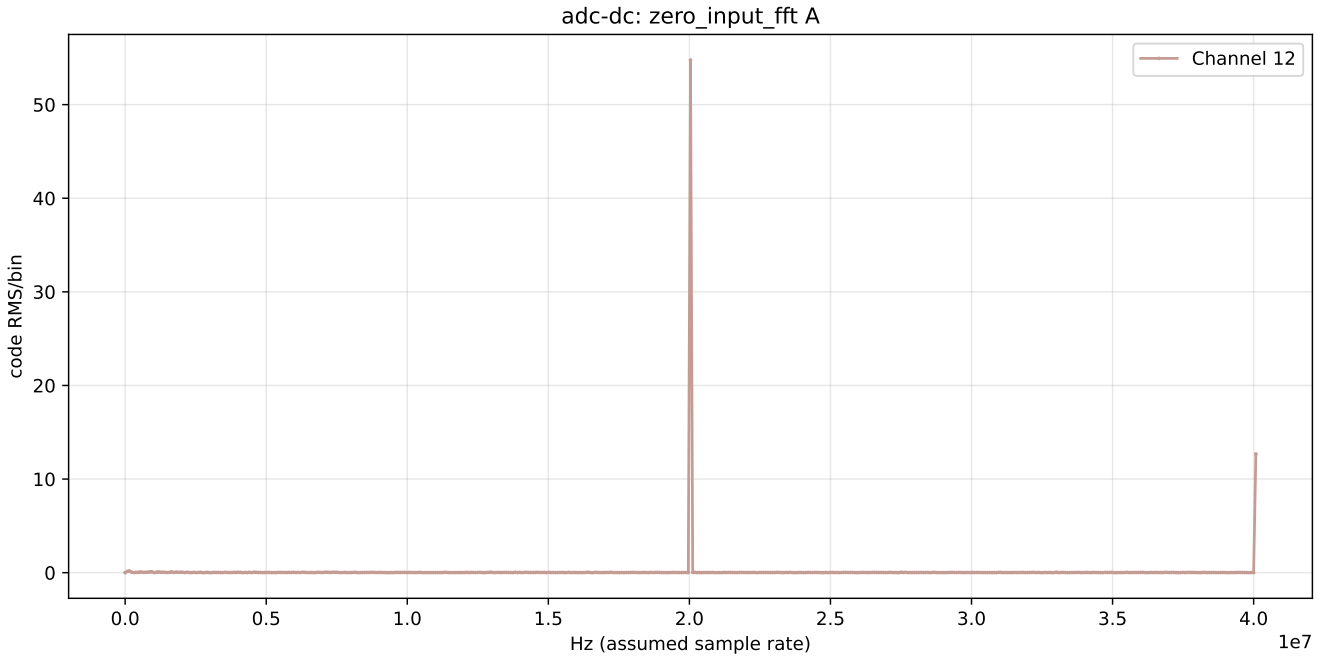
ADC total RMS includes the complete connected chain and periodic gate feedthrough. The four-phase-removed RMS removes the mean of each sample phase; it is a residual statistic, not an intrinsic ADC noise specification. FFT and time axes use the declared clock frequency, not an independent frequency measurement.

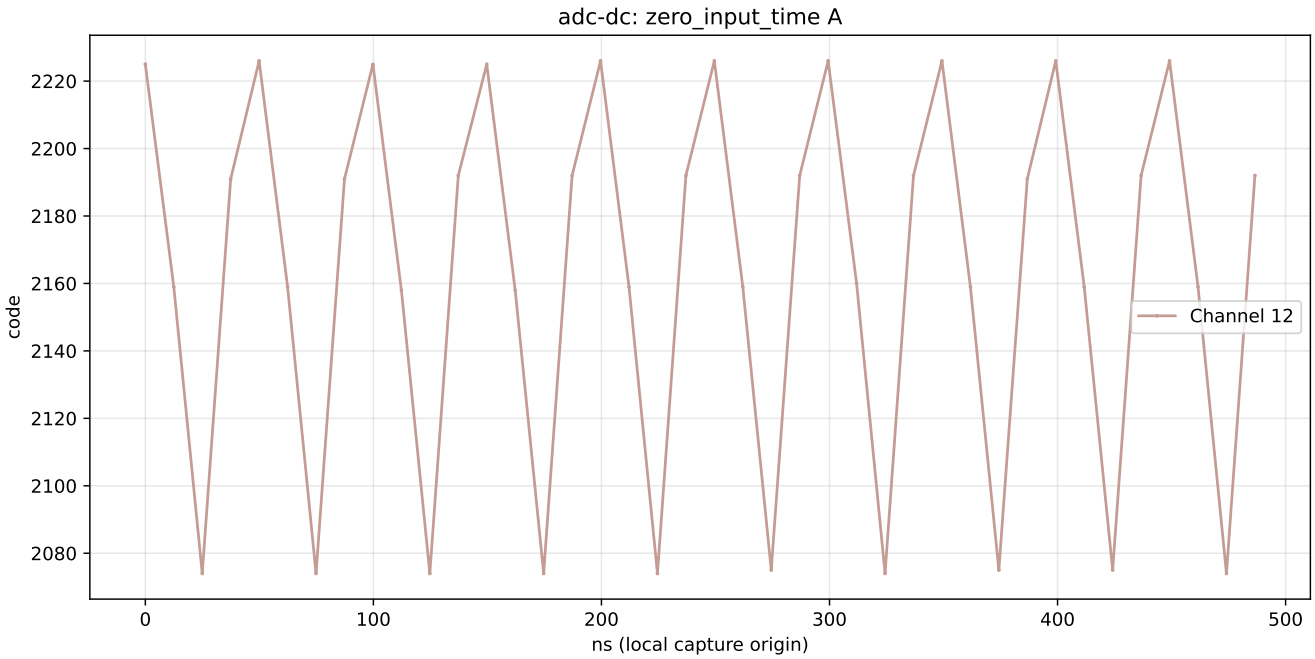
5 Measurements and common channel plots

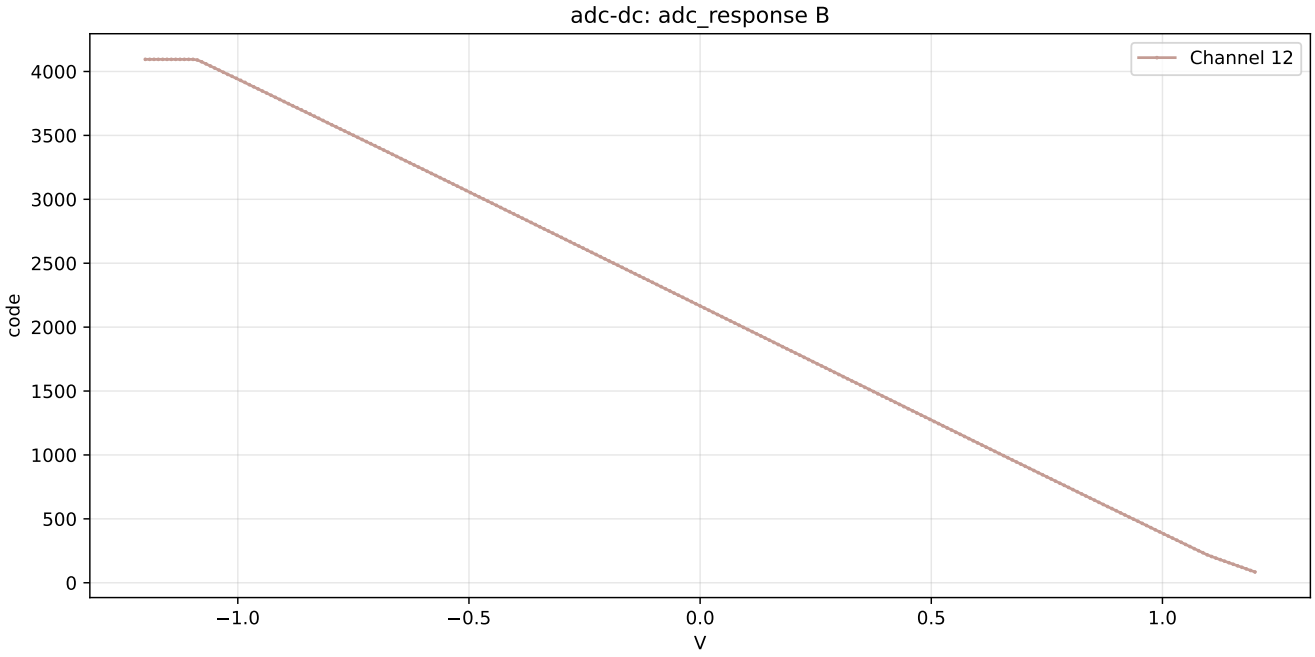
Test	Channel	Bank	Observable	Value	Unit
adc-dc	12	A	gain	-1784.205440 671235	code/V
adc-dc	12	A	zero_mean	2162.081133 9198435	code
adc-dc	12	A	zero_total_ rms	56.227545 88987761	code
adc-dc	12	A	zero_four_ phase_remove d_rms	0.592637 1250728474	code
adc-dc	12	B	gain	-1784.269486 9545635	code/V
adc-dc	12	B	zero_mean	2165.847507 3313784	code
adc-dc	12	B	zero_total_ rms	64.408946 78040004	code
adc-dc	12	B	zero_four_ phase_remove d_rms	0.562557 1866355403	code
adc-lines	12	A	adjacent_ pair_states	44	state
adc-lines	12	A	visited_codes	3147	code
adc-lines	12	B	adjacent_ pair_states	44	state
adc-lines	12	B	visited_codes	3184	code
integrator	12	A	balance_delta	0.006030 008721686399	V
integrator	12	A	gain	-2530.411960 665549	code/V
integrator	12	A	near_balanc e_change	0.130164 91045236994	code
integrator	12	B	balance_delta	0.007424 191555263186	V

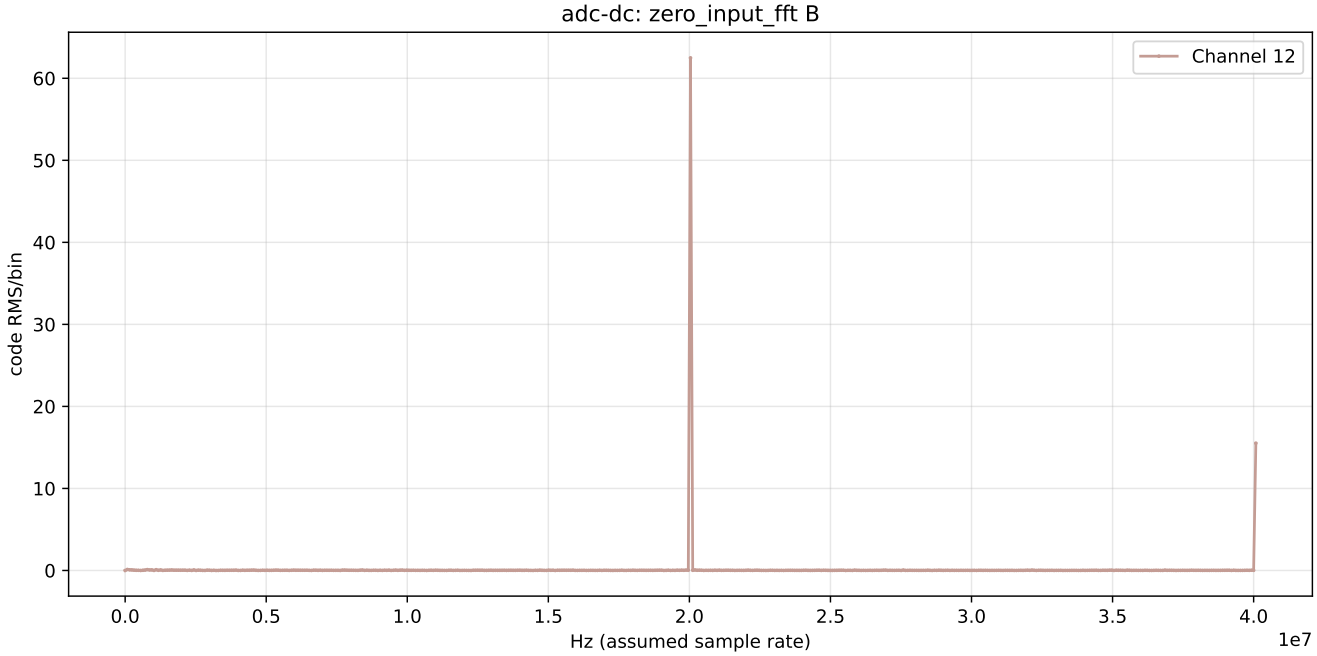
Test	Channel	Bank	Observable	Value	Unit
integrator	12	B	gain	-2448.007789 5617496	code/V
integrator	12	B	near_balance_change	0.111943 22217075807	code

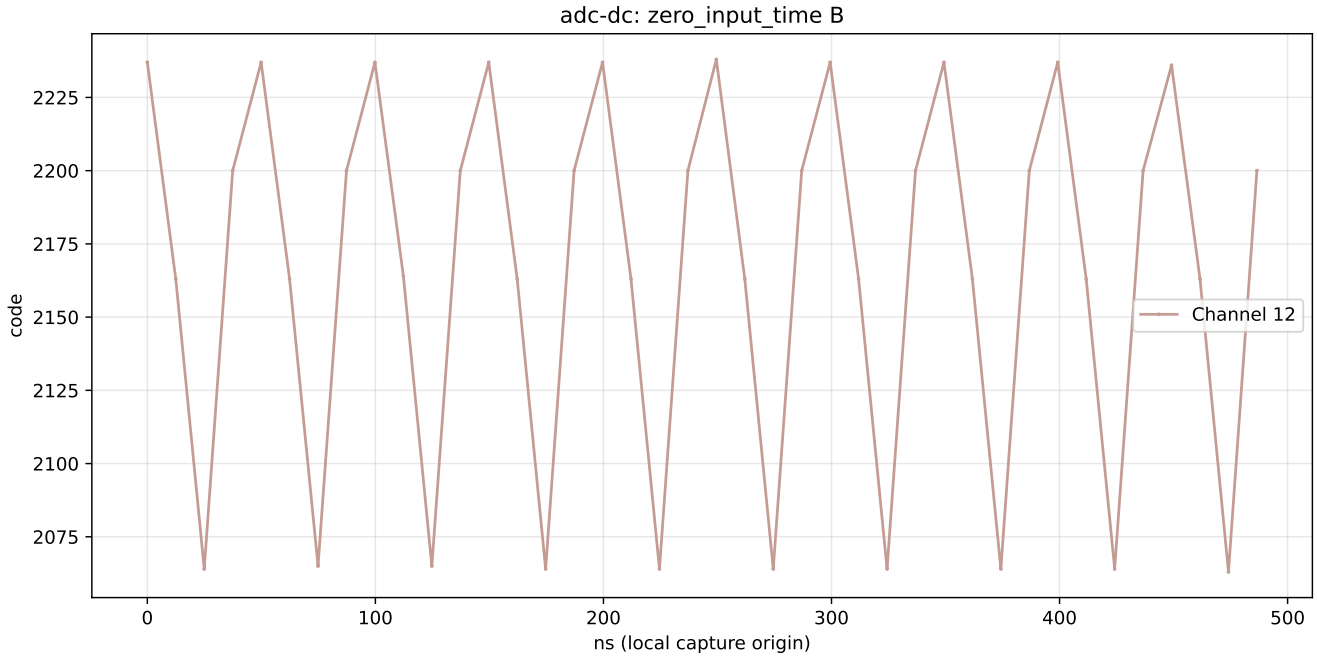


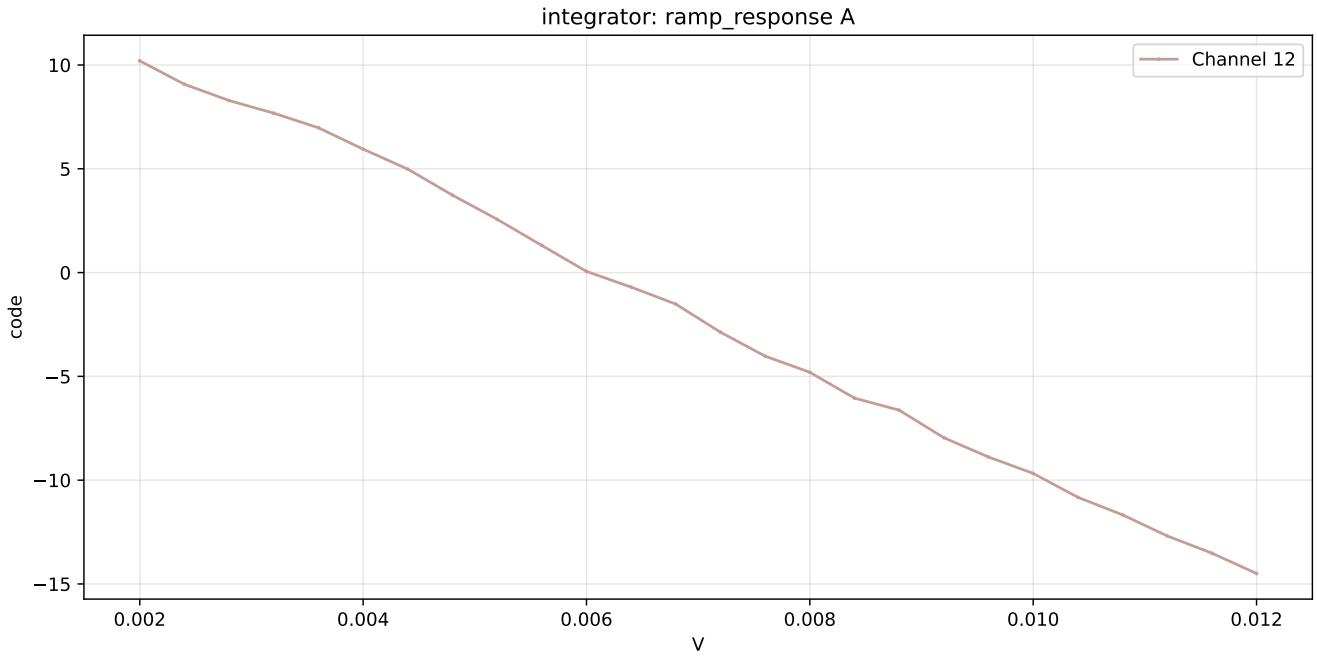


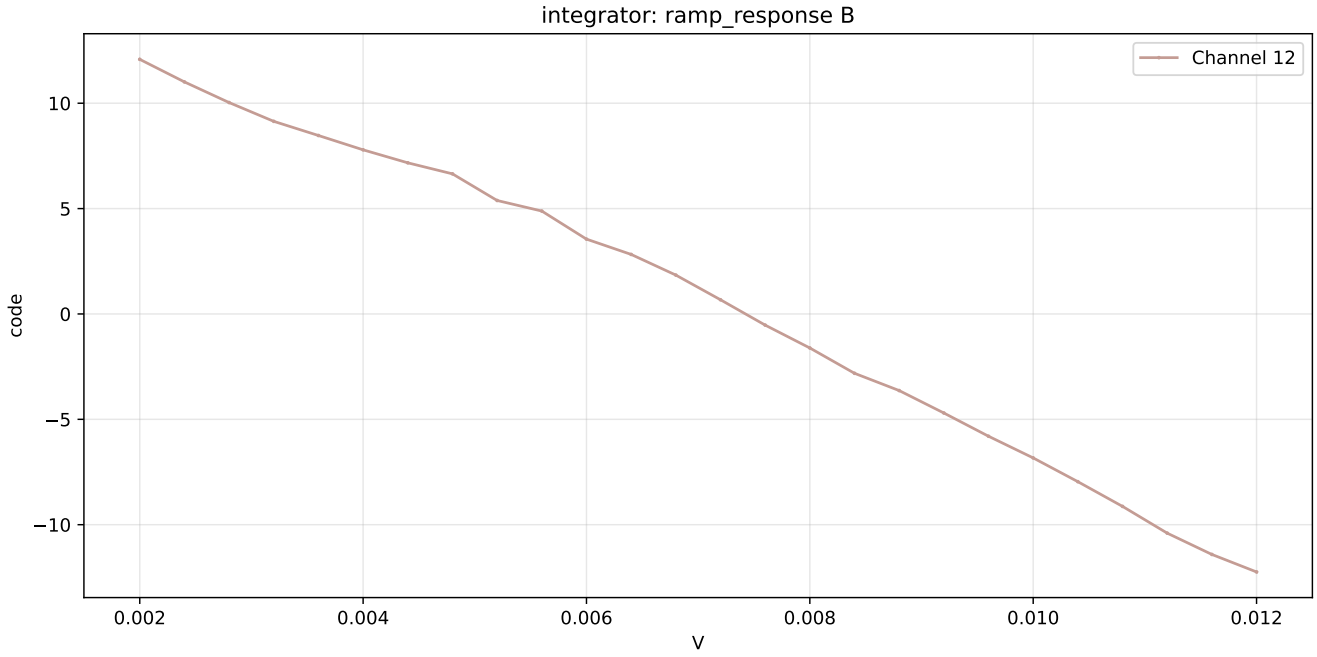

adc-dc: zero_input_time A
code
ns (local capture origin)
Channel 12
2220
2200
2180
2160
2140
2120
2100
2080
0
100
200
300
400
500

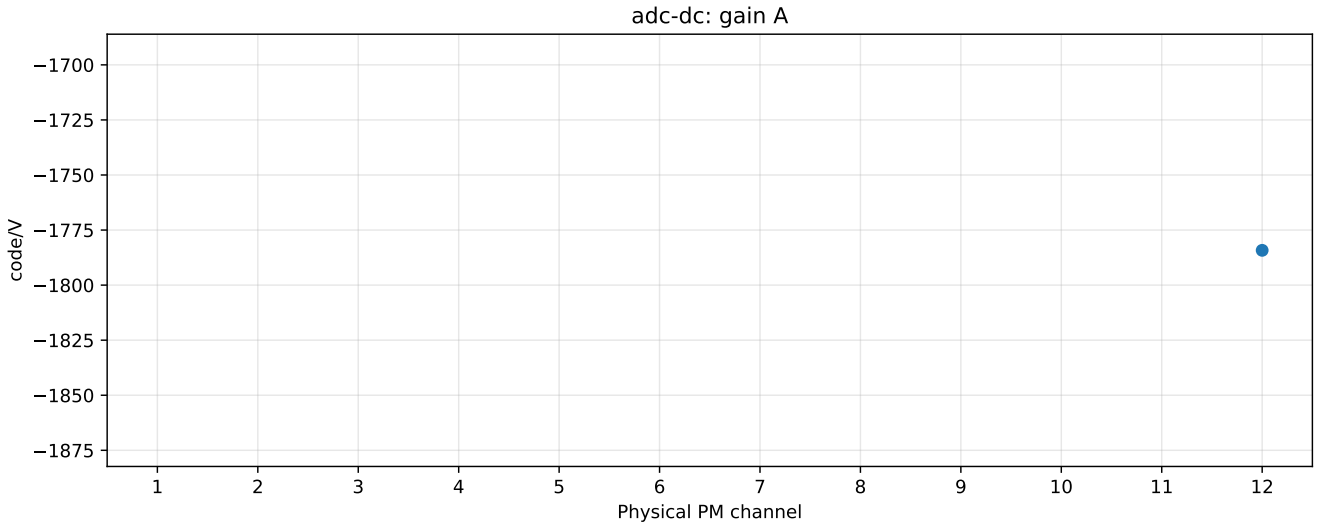

adc-dc: adc_response B
code
V
Channel 12
4000
3500
3000
2500
2000
1500
1000
500
0
−1.0
−0.5
0.0
0.5
1.0

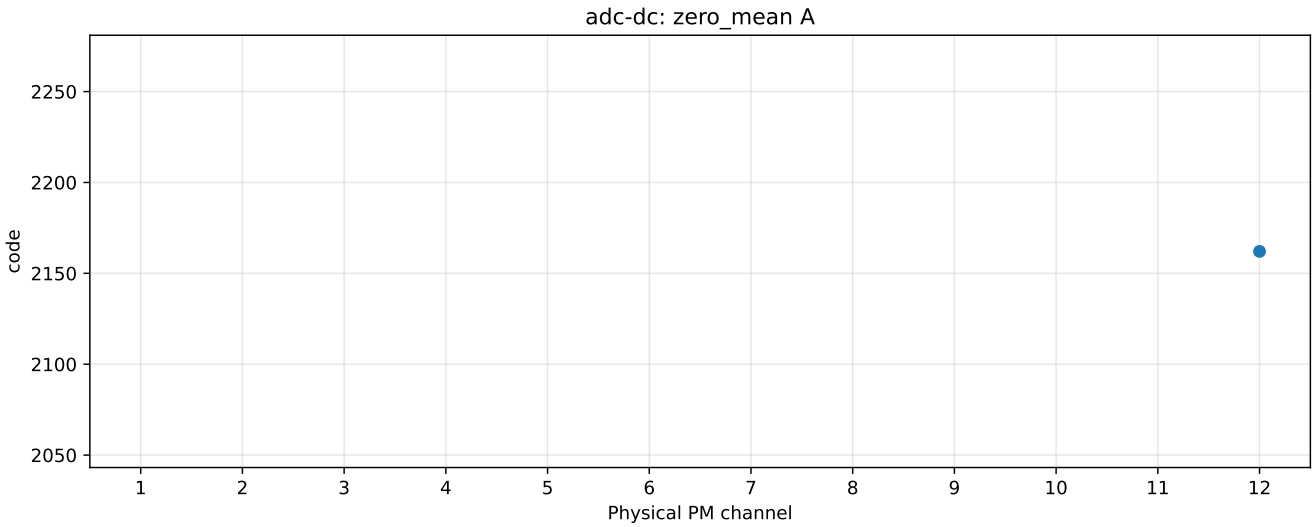

adc-dc: zero_input_fft B
Channel 12
code RMS/bin
0
10
20
30
40
50
60
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Hz (assumed sample rate)
1e7

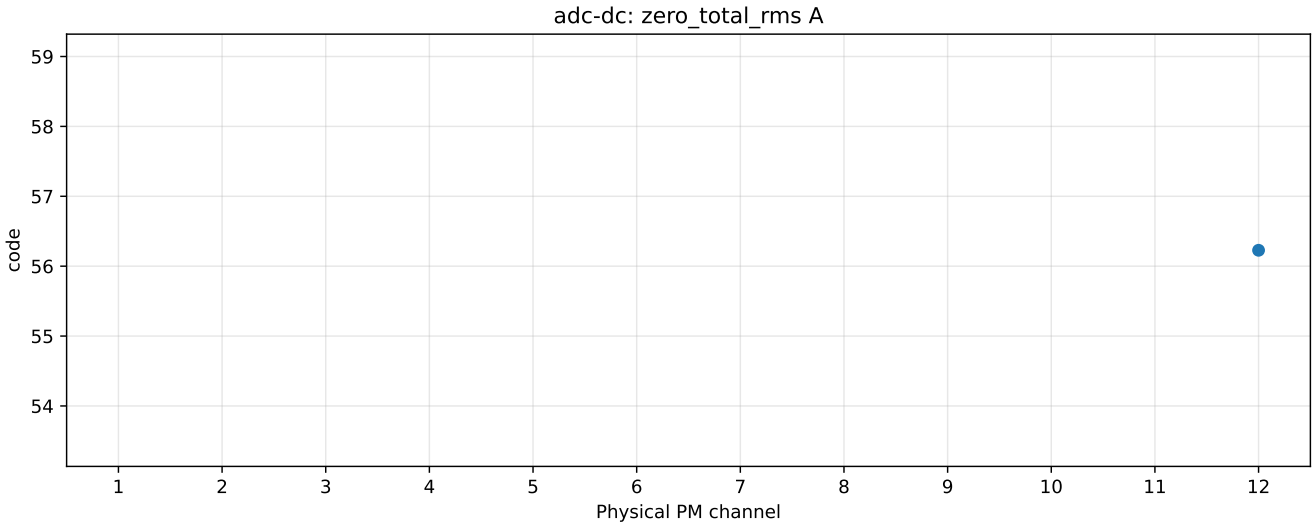

adc-dc: zero_input_time B
Channel 12
code
2075
2100
2125
2150
2175
2200
2225
0
100
200
300
400
500
ns (local capture origin)

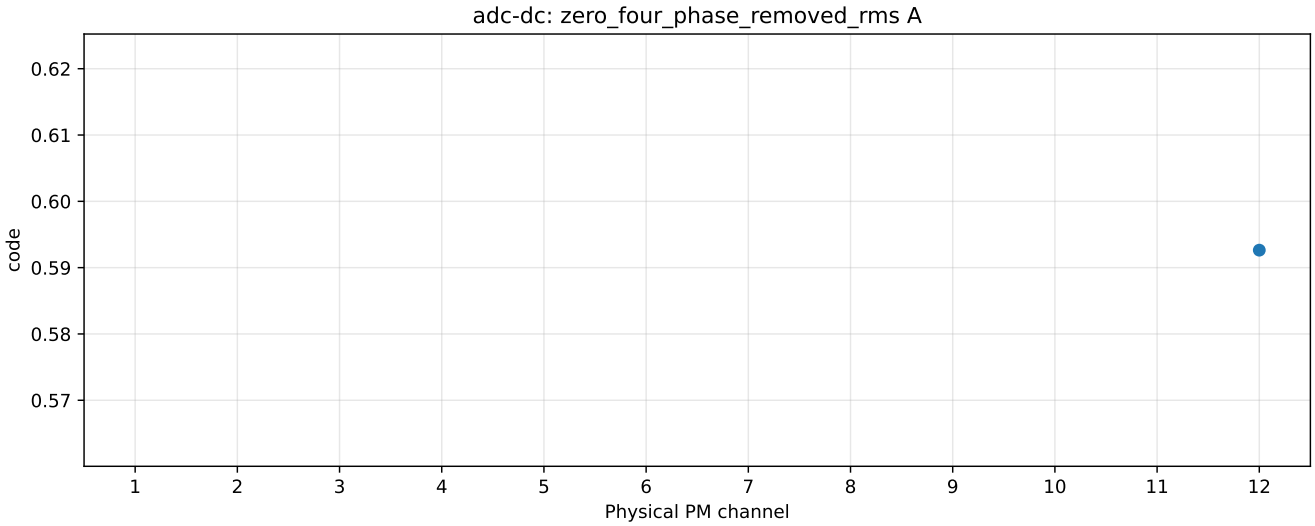

integrator: ramp_response A
Channel 12
code
V
10
5
0
−5
−10
−15
0.002
0.004
0.006
0.008
0.010
0.012

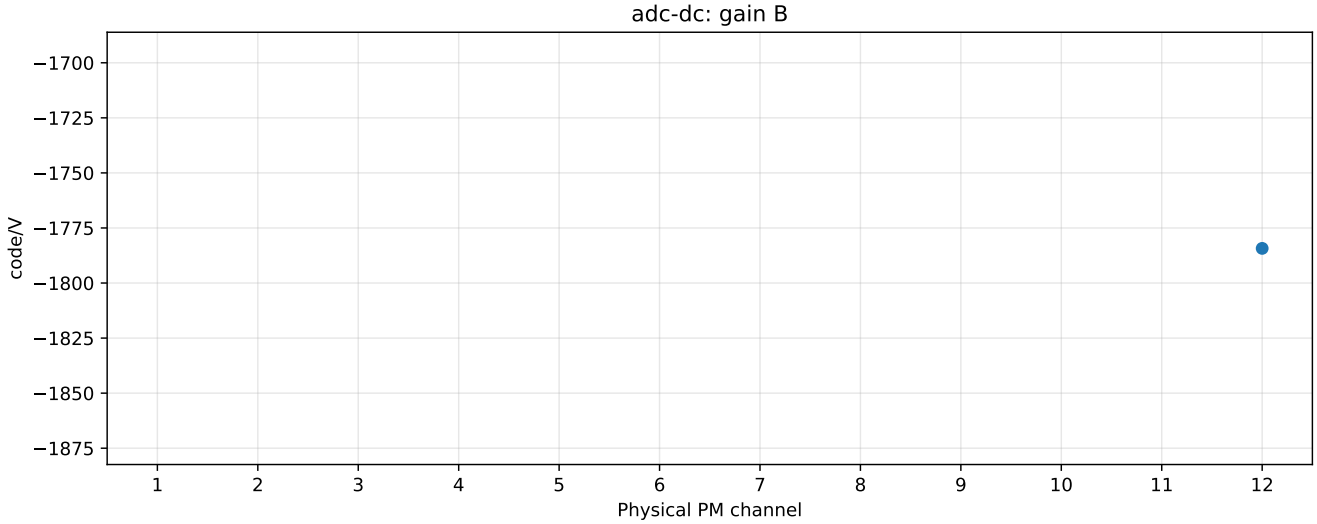

integrator: ramp_response B
Channel 12
code
V
10
5
0
−5
−10
0.002
0.004
0.006
0.008
0.010
0.012

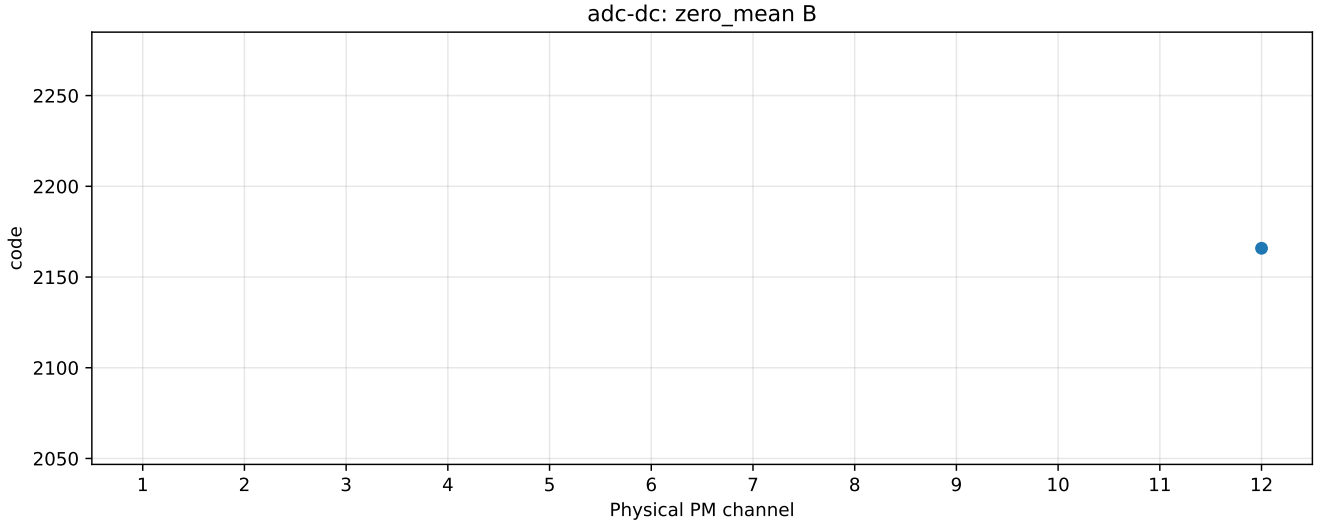

adc-dc: gain A
code/V
Physical PM channel
−1700
−1725
−1750
−1775
−1800
−1825
−1850
−1875
1
2
3
4
5
6
7
8
9
10
11
12

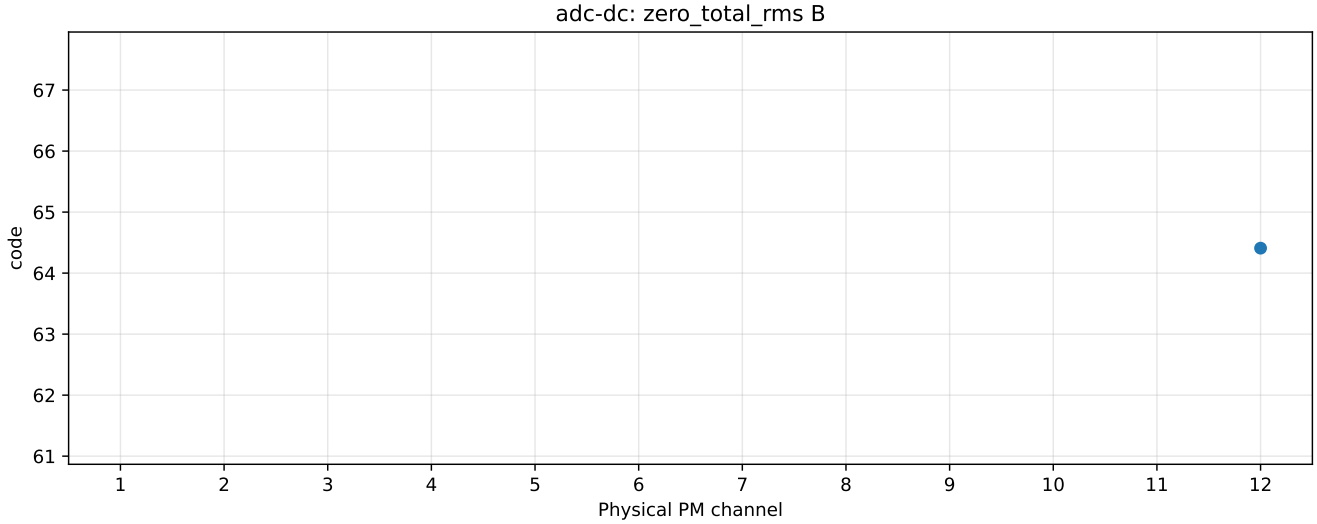

adc-dc: zero_mean A
code
2250
2200
2150
2100
2050
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

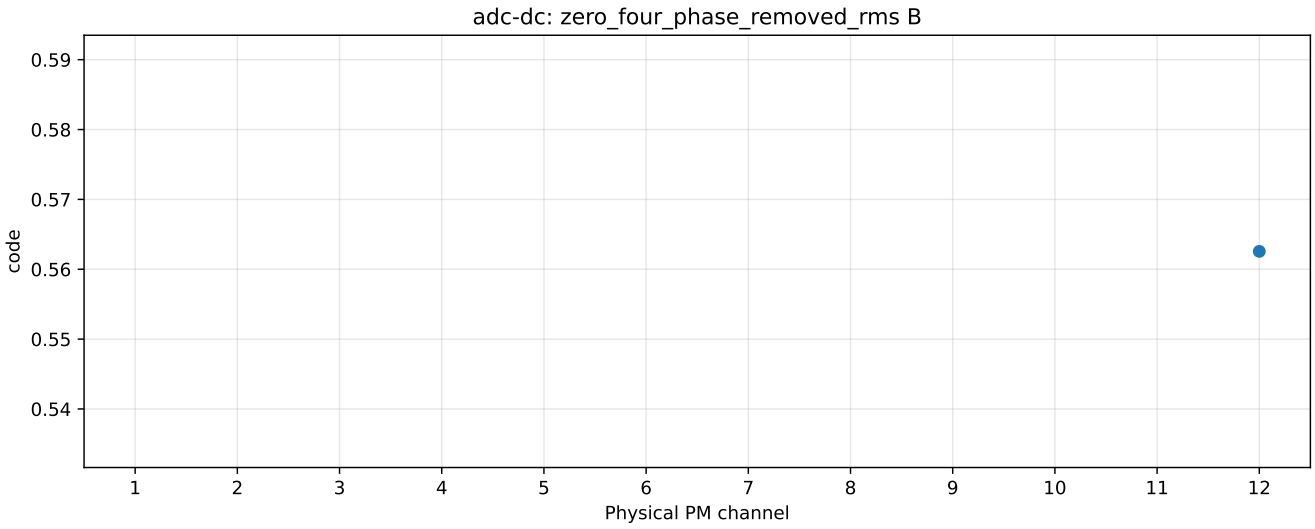

adc-dc: zero_total_rms A
code
59
58
57
56
55
54
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

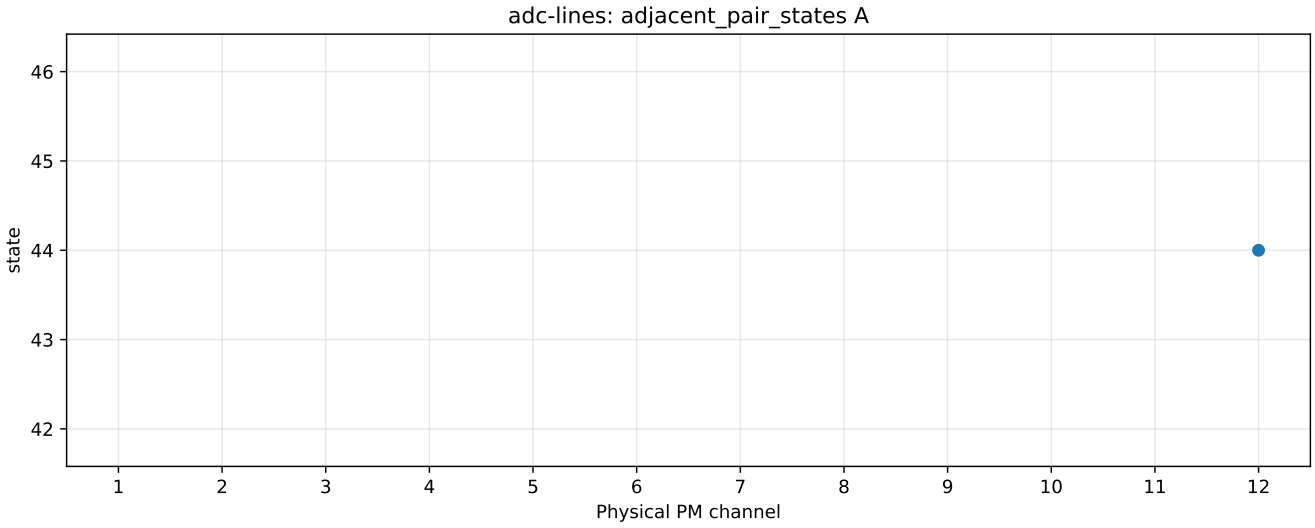

adc-dc: zero_four_phase_removed_rms A
code
0.62
0.61
0.60
0.59
0.58
0.57
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

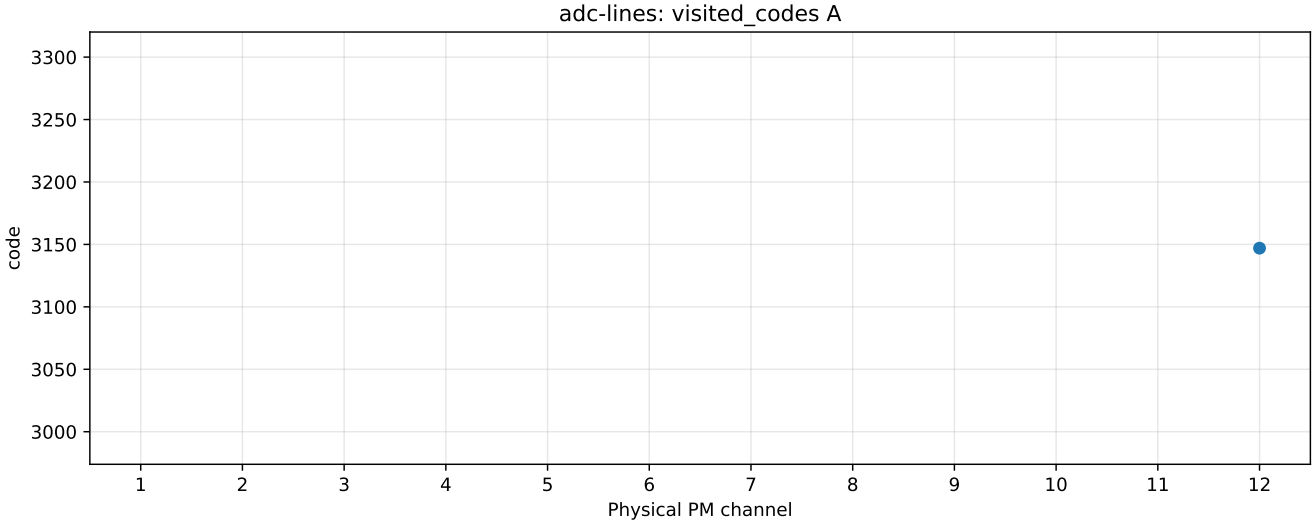

adc-dc: gain B
code/V
−1700
−1725
−1750
−1775
−1800
−1825
−1850
−1875
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

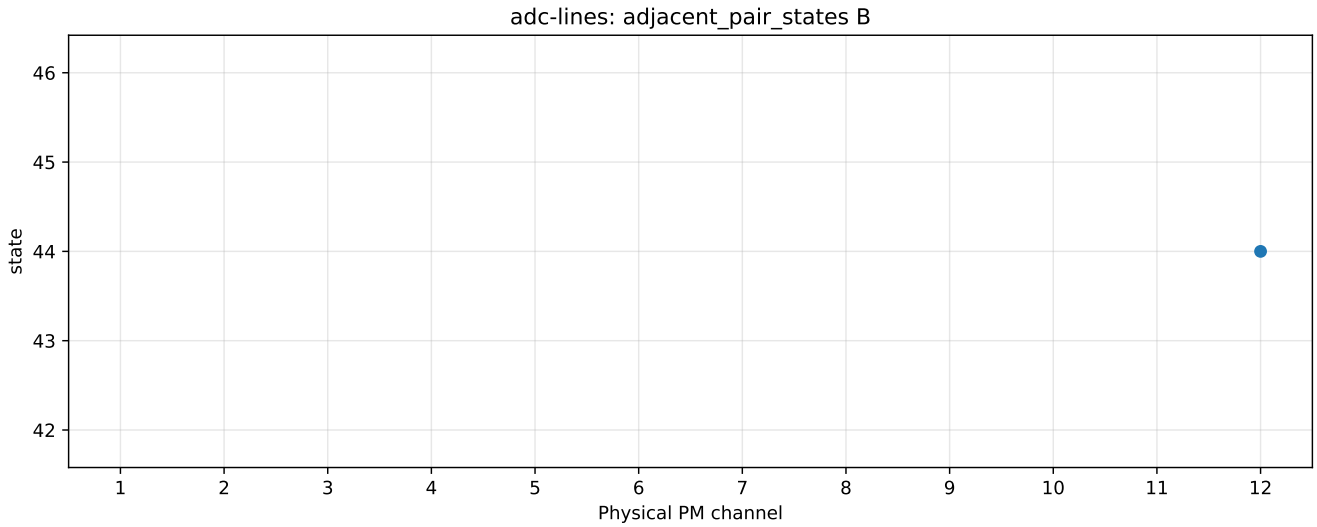

adc-dc: zero_mean B
code
2250
2200
2150
2100
2050
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

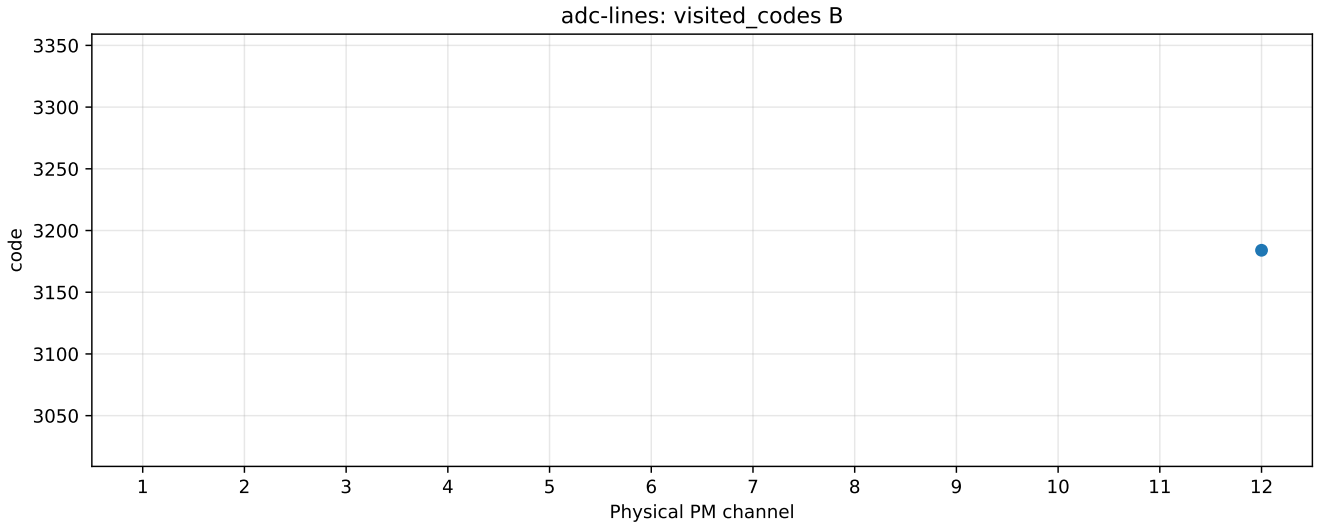

adc-dc: zero_total_rms B
code
67
66
65
64
63
62
61
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

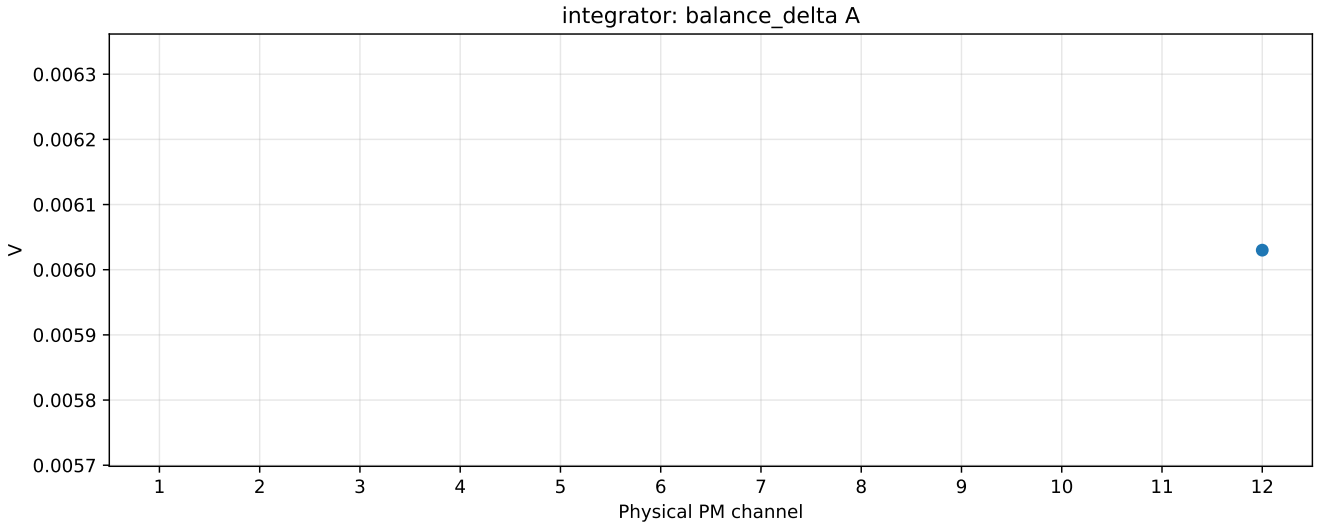

adc-dc: zero_four_phase_removed_rms B
code
0.59
0.58
0.57
0.56
0.55
0.54
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

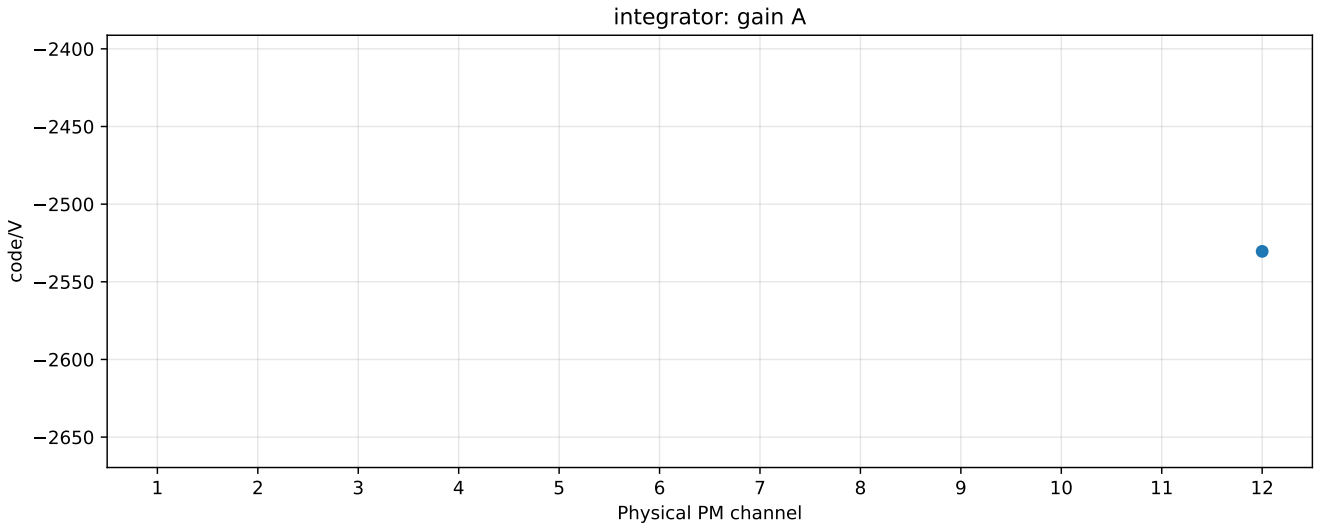

adc-lines: adjacent_pair_states A
state
46
45
44
43
42
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

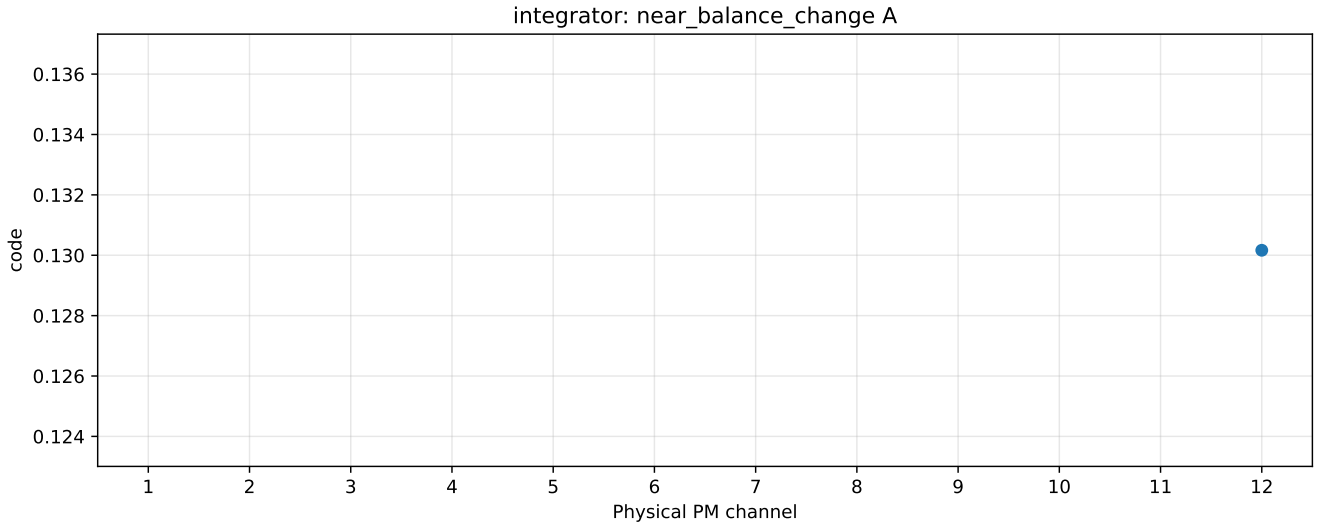

adc-lines: visited_codes A
code
3300
3250
3200
3150
3100
3050
3000
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

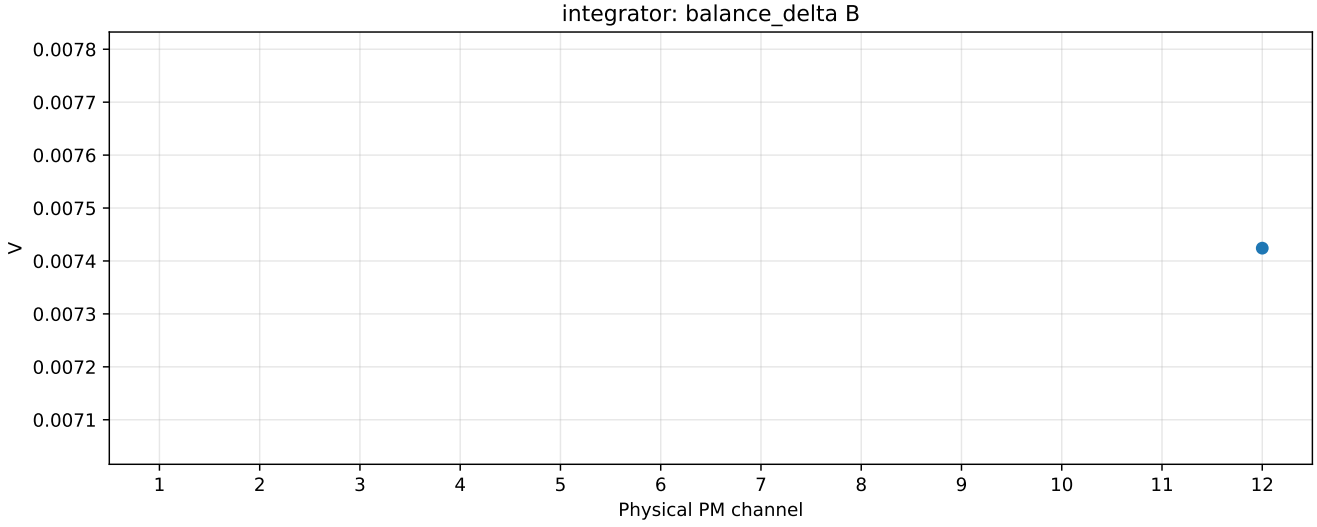

adc-lines: adjacent_pair_states B
state
46
45
44
43
42
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

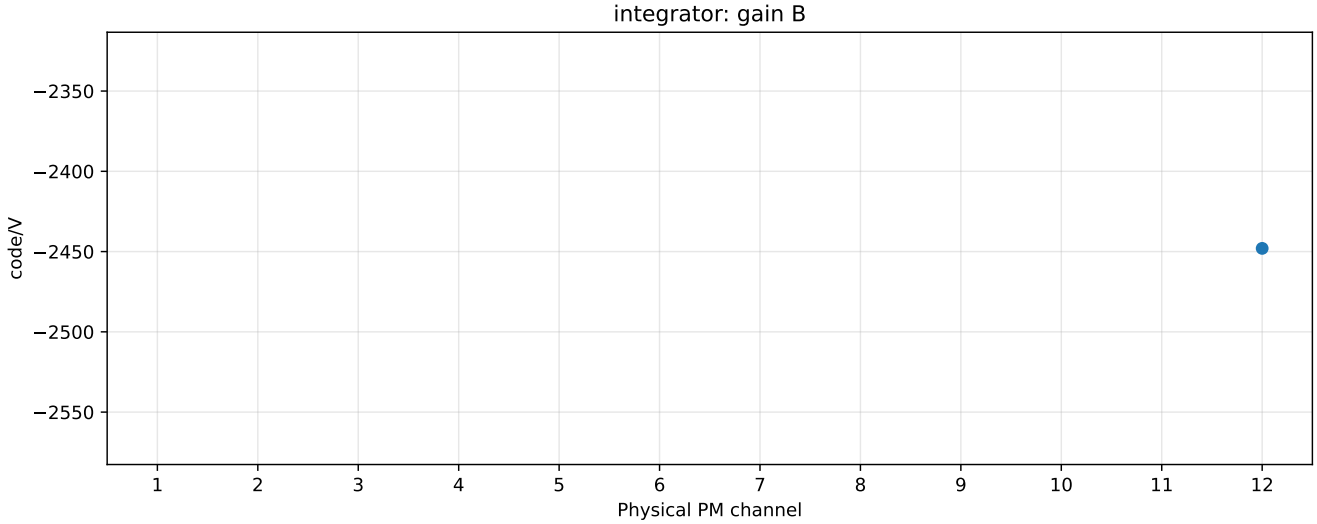

adc-lines: visited_codes B
code
3350
3300
3250
3200
3150
3100
3050
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

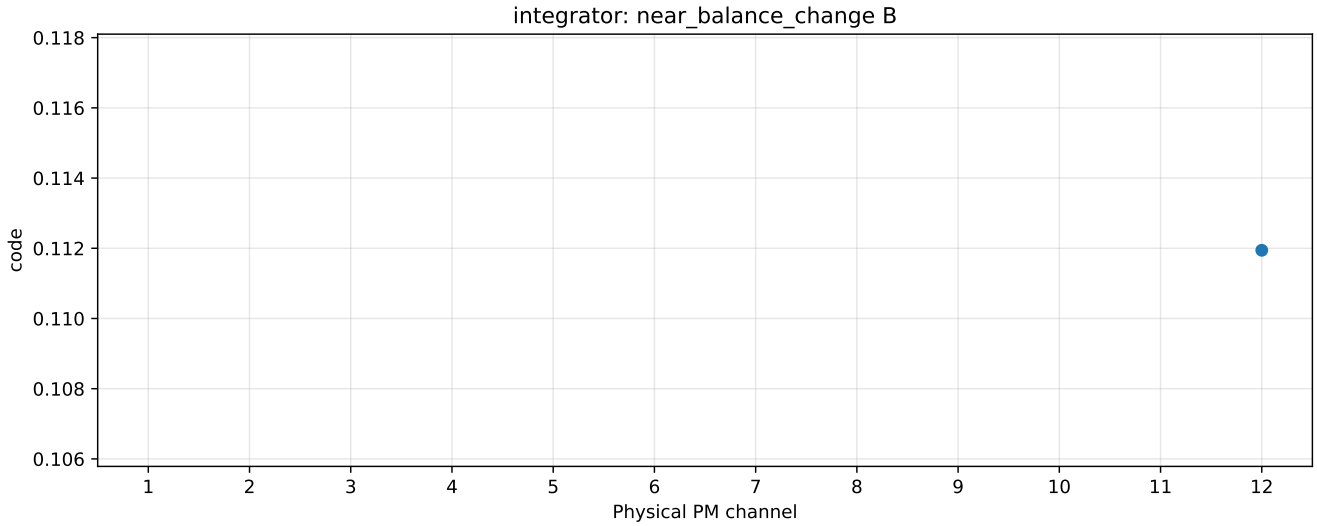

integrator: balance_delta A
V
0.0063
0.0062
0.0061
0.0060
0.0059
0.0058
0.0057
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


integrator: gain A
code/V
−2400
−2450
−2500
−2550
−2600
−2650
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


integrator: near_balance_change A
code
0.136
0.134
0.132
0.130
0.128
0.126
0.124
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


integrator: balance_delta B
V
0.0078
0.0077
0.0076
0.0075
0.0074
0.0073
0.0072
0.0071
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


integrator: gain B
code/V
−2350
−2400
−2450
−2500
−2550
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


integrator: near_balance_change B
code
0.118
0.116
0.114
0.112
0.110
0.108
0.106
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel